WASP-50b

R-0884

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_250930` · created 2026-07-29T01:02:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460948.71 +/- 0.13 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.05
Transit depth (Rp/Rs)^2	2.66 +/- 1.63 [%]
Orbital Inclination (inc)	86.67 +/- 2.89
Airmass coefficient 1 (a1)	0.123 +/- 0.086
Airmass coefficient 2 (a2)	0.7 +/- 0.21
Scatter in the residuals of the lightcurve fit is	79.84 %
Best Comparison Star	#2 - [281, 209]
Optimal Aperture	1.22
Optimal Annulus	3.5
Transit Duration (day)	0.022 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.163 ± 0.05 vs 0.1404 (deviation 0.45σ)
Duration fit vs published	0.022 d vs 0.0754 d (deviation 1.84σ)
Mid-time O–C	-135.1 min
Depth SNR	3.3
Residual scatter	79.84 %

Light curve

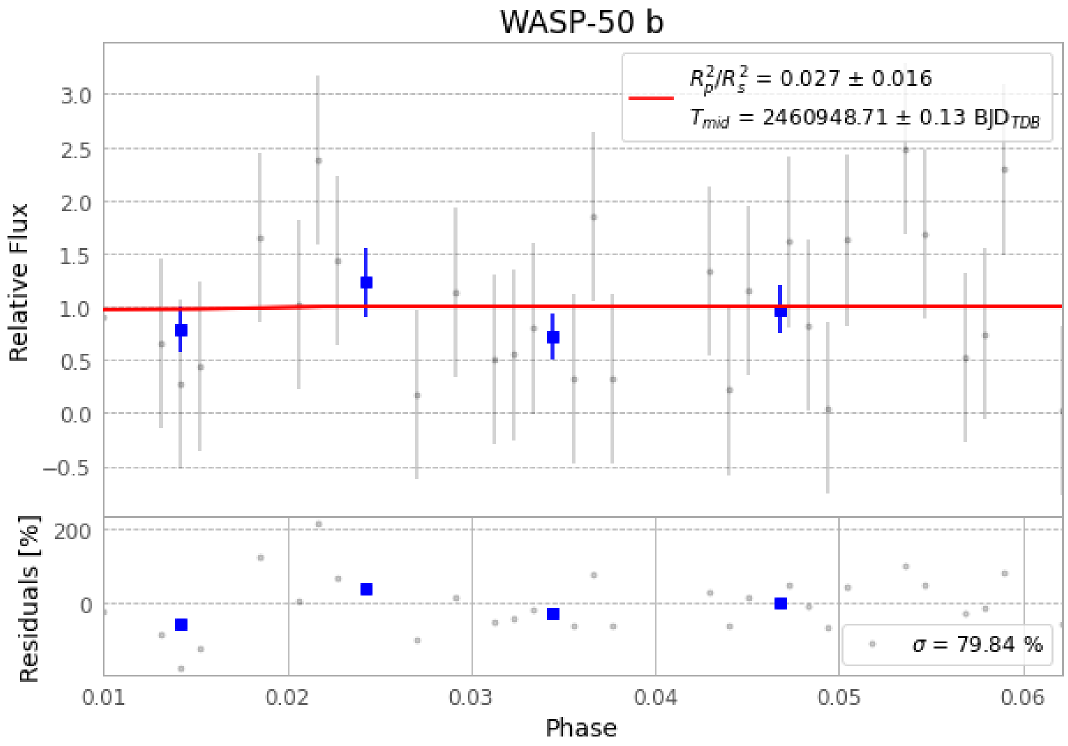

Figure 1 — FinalLightCurve_WASP-50 b_2025-09-29.png (SHA-256 5474ec47cd3c679f…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [237, 446], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 14a849fea57eca68…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

73 FITS frames (48 MB), `WASP-50250930051515.FITS` → `WASP-50250930085116.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-250930.log (a828b127475f…), FinalLightCurve_WASP-50 b_2025-09-29.png (5474ec47cd3c…), FinalLightCurve_WASP-50 b_2025-09-29.csv (58dc949e2ce1…)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)